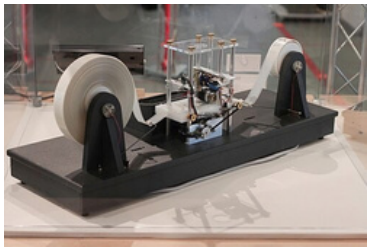
IA & IA GENERATIVE

IA as a definition (from my prespective)

ia is like a bot. advanced developpers created in the purpose to help and facilates their work and ofcourse out of curiosity.
if we look back to years before the human itself had always tried to make robotic things a robot who thinks and could create somthing out of nothing.so let me take to 1950 where the famous alan turning decided its time to change the cycle and start creating and discovering the Computing Machinery and Intelligence.

The start of lets say IA



Turing machine

Alan turning a british mathematic he formalised the concepts of algorithm and computation with the Turing machine
He also contributed the Turing test to philosophy of artificial intelligence and the Turing pattern concept to mathematical and theoretical biology. In other terms he may did create the idea of IA

And after that a lot of other genius and humans got deeper like 1955: John McCarthy(the one who got the credit or as we call him "The Father of AI".1958: Frank Rosenblatt....

The exact timeline of the developpements

(credits goes to gemini in this section)

- 1997 (The Early Transition): IBM's Deep Blue defeated chess champion Garry Kasparov. While it still relied heavily on human-coded rules and brute-force computing, it proved that specialized data processing could defeat human intuition.
- Early 2000s (The Rise of Big Data): The explosion of the internet provided the massive datasets (millions of images, billions of words) required to train modern data-driven AI systems.
- 2009 (The GPU Breakthrough): Researchers, including Andrew Ng, realized that Nvidia graphics cards (GPUs) could process the heavy math of neural networks 10 to 100 times faster than normal computer processors (CPUs).
- 2012 (The Modern Era Begins): AlexNet won the ImageNet competition using a combination of deep neural networks, millions of web images, and GPUs. This officially proved that data-driven "Deep Learning" completely outperformed old rule-based coding, changing AI forever

The IA & the IA generative (from my prespective)

Artificial Intelligence (AI / IA): is a coded bot creates content based on an already existing data to simplify a human writes their own prompet (prompet is a text where we tend to give and order to the ia)and the ia tend to use the keywords we add so that he gives us an exact response based on wht the code that created him have

Generative AI (IA Générative): Its like a bot who is made to create new original content sush as videos,pictures,codes,texts,overall it tries to generate content that could echo the human's output to simplify an intellegent bot like a magic box you dont know wht the answer could be because he will created the min you put ur prompet

The Structural Hierarchy

(credits goes to gemini in this section)

- IA (Intelligence Artificielle) $\rightarrow$ General:
  Definition: The broad field of computer science focused on creating machines capable of mimicking human intelligence to perform general tasks.
- ML (Machine Learning):
  Definition: A subset of AI where systems learn from data to make decisions or predictions without being explicitly programmed.
  Class (Classification): Categorizing data into distinct groups (e.g., separating emails into spam or not spam).
  Prediction: Foretelling numeric values or future trends based on historical patterns.
- DL (Deep Learning) $\rightarrow$ Réseaux de Neurones:
  Definition: A specialized subfield of Machine Learning that uses multi-layered Neural Networks (inspired by the human brain) to process complex patterns in large datasets.
- LLM (Large Language Models):
  Definition: Deep Learning models trained on massive amounts of text data to understand and generate human language.
  Création Text, Video, Audio: Instead of just analyzing data, LLMs focus on generating original content across formats like text, video, and sound.

Limits of Ia & Ia generative

Im going to break it throw in 2 point:
1-hallucination: whish is when the ia gives us factually false, inaccurate, or fabricated responses
2-Biais: when the outcome is unfair not what we wanted to explain its when they get the wrong keywords from ou prompet whish it all end up giving us some unwanted information error in the database